CMPT 459 Data Mining - Fall 2025

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Course Project Guidelines

1. Objective

In this project, student groups will explore a dataset of their choice and apply data mining and data analysis techniques to develop a deeper understanding of the data mining process. This year, the course project will consist of three main deliverables: **Project Proposal, Data preprocessing and exploratory data analysis (EDA), Final Project Delivery.**

You must use a **Git repository** to submit your project. The SFU GitHub server is a good way to get one. Group members must commit their own contributions to the repo. Please give the instructor and TAs (ester, dad13, akhoeini) permission as collaborators on the repository.

You are encouraged to publicize and open-source your work on GitHub or similar.

Note: There will be **no presentations** required for this year's course project.

2. Project Proposal

Each group should consist of **2–3 students**. You are encouraged to coordinate with classmates to form groups early. Students not in a group by **September 23** will be randomly assigned. By Tuesday, **September 30, 2025**, submit a brief proposal **(≤1 page**). The proposal should contain

- The name of and a link to your chosen dataset.
- Statistics of the dataset, including the numbers of samples and features.
- A short rationale for choosing the dataset.
- A list of questions that you want to investigate using that dataset.

Dataset Requirements

- **Minimum size**: 1,000 samples
- **Minimum features**: 10 or more features
- **Source**: Public repositories such as **Kaggle**, **UCI Machine Learning Repository**, or similar platforms

- **Feature composition**: Must contain a mix of **numerical** and **categorical** features
- **Target variable**: Must include at least **one target variable** suitable for classification tasks.

3. Data Preprocessing & Exploratory Data Analysis (EDA)

Perform exploratory data analysis to understand the structure and distribution of the dataset. Then preprocess the dataset to ensure it is clean and ready for analysis. This should include handling missing values, encoding categorical variables, and normalizing numerical features.

Exploratory Data Analysis (EDA) Requirements

- Perform basic EDA to understand the structure and distribution of the dataset.
- Plot distributions of key features using **histograms**, box plots, etc.
- Visualize relationships between features and identify correlations using **heatmaps**.
- Discuss key insights drawn from EDA and potential challenges with the dataset (e.g., class imbalance, highly correlated features).

Data Preprocessing Requirements

- Handle **missing values** appropriately (e.g., imputation, removal).
- **Normalize/standardize** numerical features and perform **encoding** for categorical variables (one-hot encoding or label encoding).
- Perform **data augmentation** if applicable (e.g., generating synthetic samples for imbalanced datasets).
- Use **dimensionality reduction** techniques like **PCA** or **t-SNE** if the dataset has high dimensionality.

4. Final Project Delivery

The final project delivery encompasses comprehensive data mining analysis including clustering, outlier detection, feature selection, classification, and model optimization. Apply multiple techniques and provide a thorough analysis of your results.

Cluster Analysis

- Apply **at least one clustering algorithm per team member** (e.g., K-Means, DBSCAN, Hierarchical Clustering).

- Visualize clustering results using methods like **PCA** or **t-SNE** for dimensionality reduction, followed by 2D or 3D scatter plots.
- Evaluate the clustering performance using metrics such as **Silhouette Score**, **Calinski-Harabasz Index**, or **Davies-Bouldin Index**.
- Discuss the appropriateness of the clustering algorithms for your dataset and compare their performances.

Outlier Detection

- Perform at least one **outlier detection** per team member, using techniques like **Isolation Forest, Local Outlier Factor (LOF),** or **Elliptic Envelope**.
- Visualize the outliers using plots (e.g., scatter plots with outliers marked).
- Analyze the outliers: Are they noise, or do they contain important information? Decide whether to keep or remove them for further analysis

Feature Selection

- Apply **feature selection** techniques such **as Recursive Feature Elimination (RFE), Lasso Regression**, or **mutual information** to reduce dimensionality.
- Discuss the importance of selected features and their impact on the classification task.
- Evaluate the model with and without feature selection to compare performance and computational efficiency.

Classification

- Use **at least one classification algorithm** per team member, such as **Random Forest, Support Vector Machines (SVM), k-NN.**
- Split the dataset into training and testing sets (e.g., 80% training, 20% testing) and perform **cross-validation** (e.g., 5-fold or 10-fold) to evaluate model consistency.
- Evaluate the models using various metrics, including:
    - **Accuracy**
    - **Precision**
    - **Recall**
    - **F1-score**
    - **AUC-ROC** for binary classification problems.
- Visualize results using **a confusion matrix** and **ROC curves**.

Hyperparameter Tuning

- Perform **hyperparameter tuning** for at **EACH** classifier using **Grid Search** or **Random Search**.

- Compare the performance of the model before and after tuning. Discuss the impact of tuning on model performance.

Final Deliverables

For the final project delivery, you must submit the following:

- A short report (**maximum 2 pages**) that includes:
    - Summary of methodology applied.
    - Key results and findings
    - Domain-specific insights derived from the analysis
    - Challenges encountered and how they were addressed.

- Jupyter notebook (**primary report**):
    - **One annotated .ipynb per group** that documents the end-to-end workflow (code cells, figures, metrics, and brief interpretations for each task).
    - Cross-reference any helper modules/scripts used in the repo.
    - Students may additionally create an individual portfolio version derived from the group notebook (optional).
- Code submission:
    - Clean, well-documented Python code in a **GitHub repository**.
    - Code should be modular, with clear sections for each task.
    - README (not a report): setup instructions, environment details, how to run the notebook/code, repository map, and data access notes.